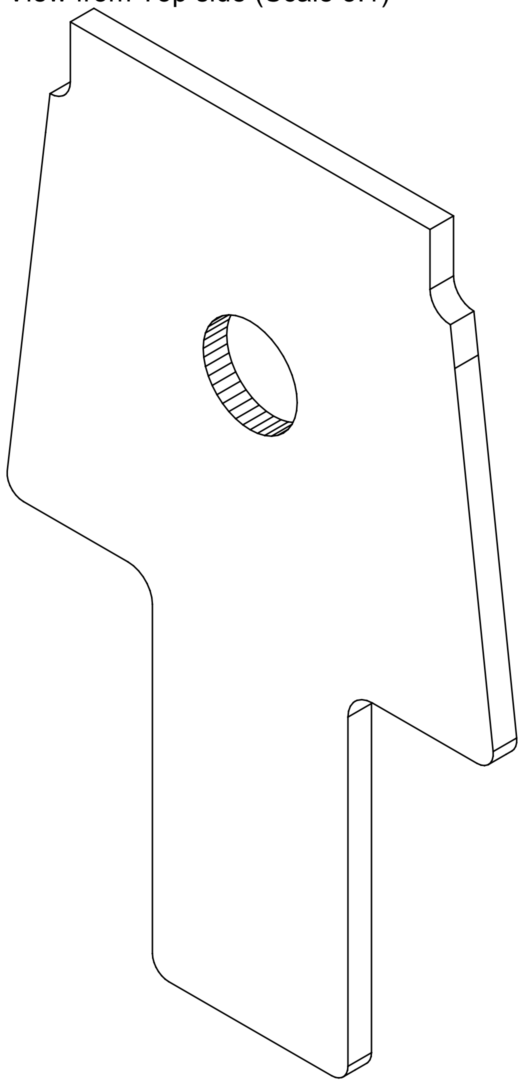
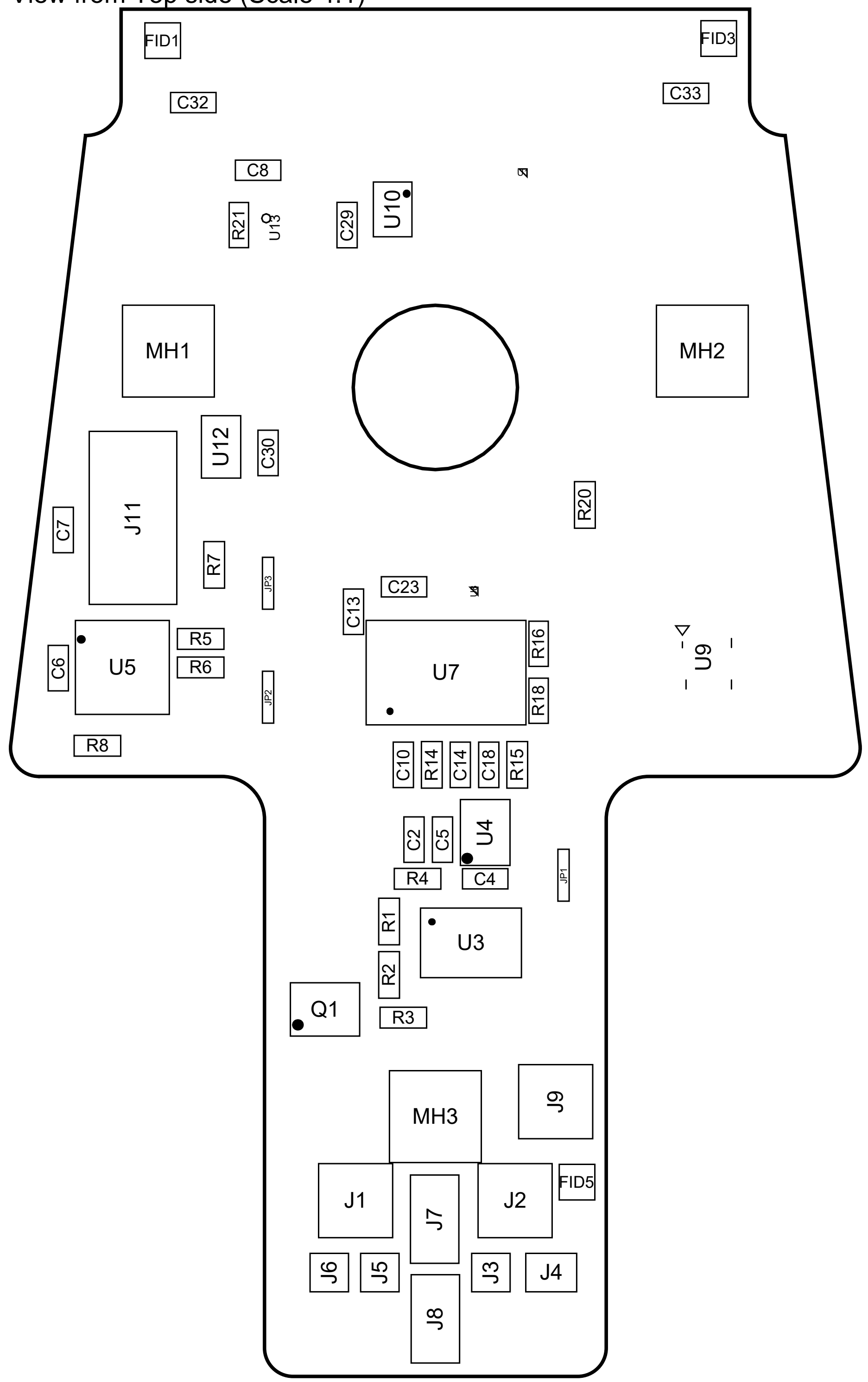
Notes:

1. Ref Des. J10 Leads have a Max Height Requirement. Need to identify length.
2.

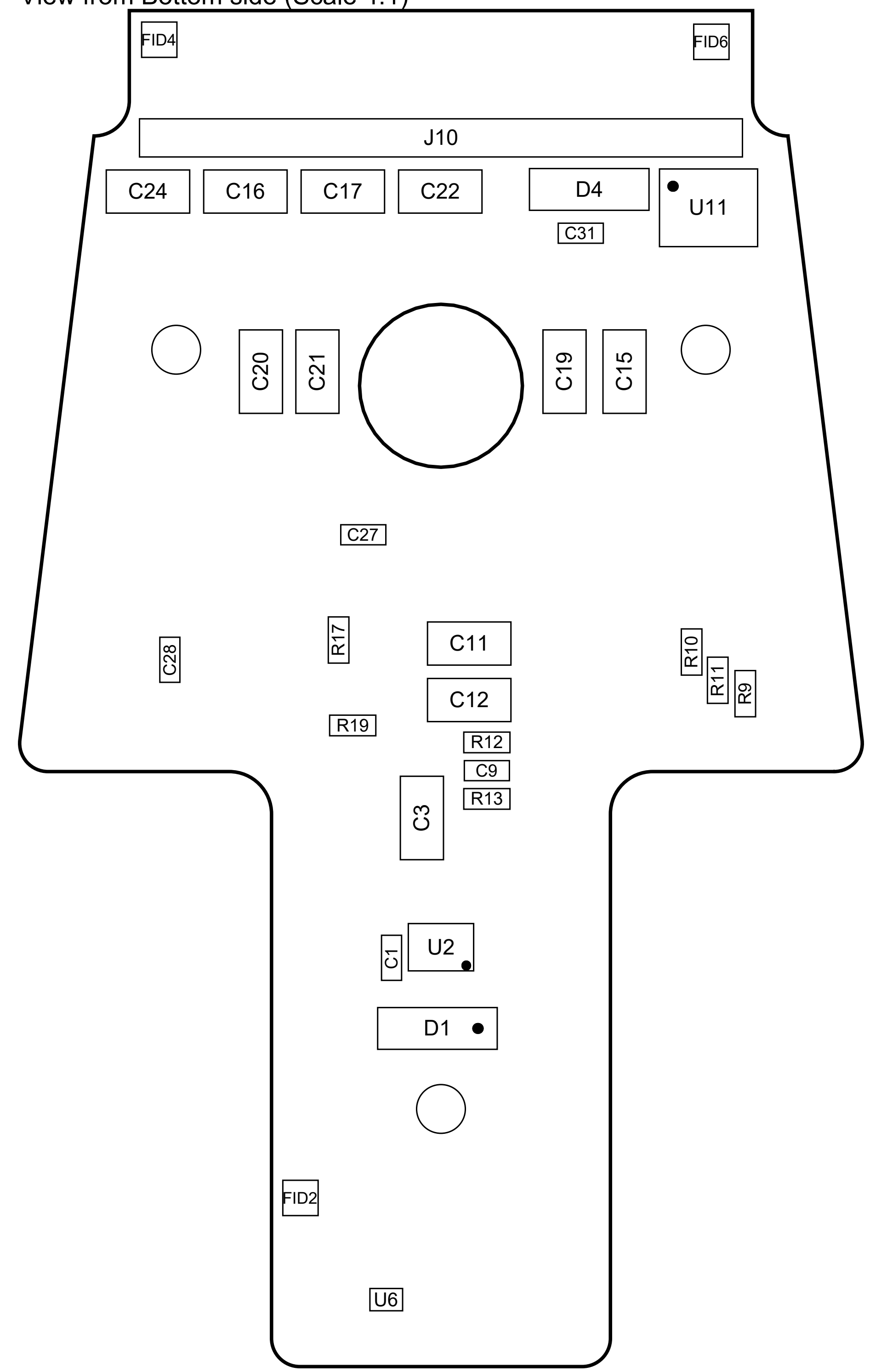
View from Top side (Scale 3:1)



View from Top side (Scale 4:1)



View from Bottom side (Scale 4:1)



REVISIONS

REV	DESCRIPTION	DATE(YEAR-MO-DA)	APPROVED
1			

PART NO. PB-F-0210

DISTRIBUTION STATEMENT A. APPROVED FOR PUBLIC RELEASE; DISTRIBUTION IS UNLIMITED.

PMIC

MECHANICAL PROPERTIES

YS
TS
EL2
RA
BH

NEXT ASSY	USED ON
=NextAssy3	=UsedOn3
=NextAssy2	=UsedOn2
=NextAssy1	=UsedOn1

APPLICATION

DO NOT SCALE DRAWING

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL±
ANGULAR: MACH± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±

THIRD ANGLE PROJECTION

MATL ENGR | MODELED BY

CONTRACT NUMBER

CONTRACTOR

DRAWN BY =DrawnBy	DATE (YEAR-MO-DA) =DrawnDate
CHECKER	ENGINEER 1
ENGINEER	QUALITY ENGINEER =QA

DRAWING APPROVAL

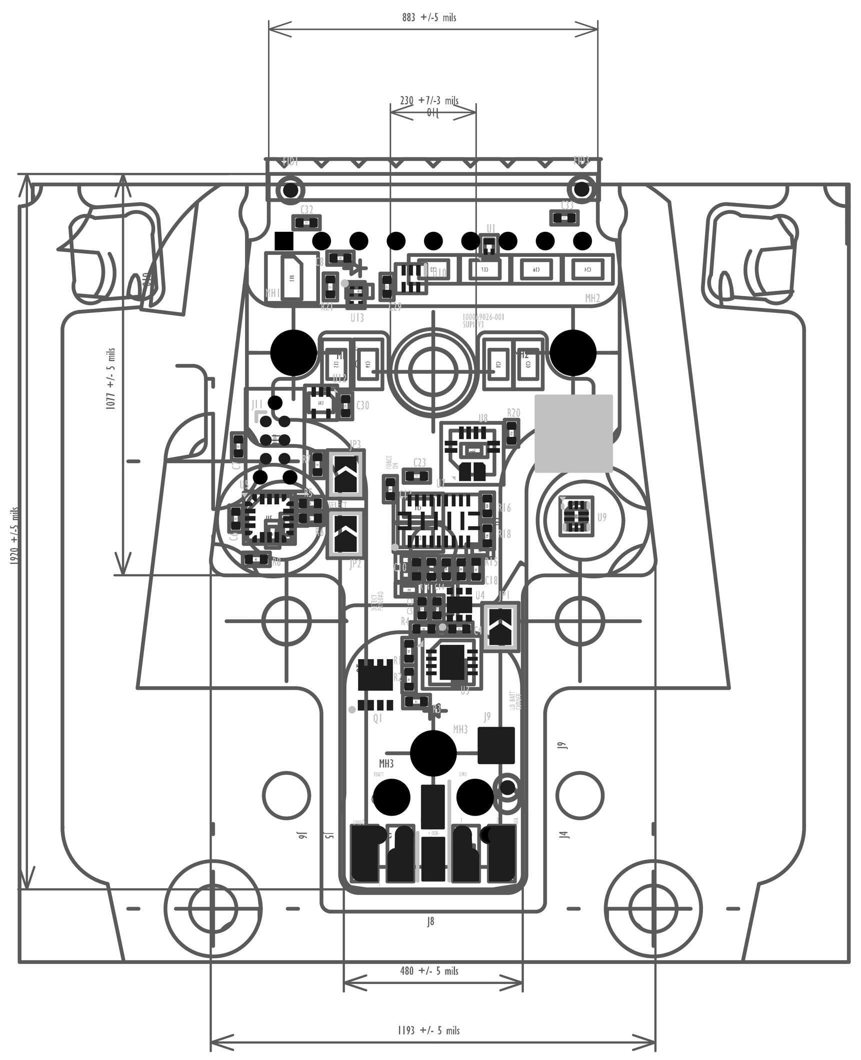
DESIGN APPROVAL

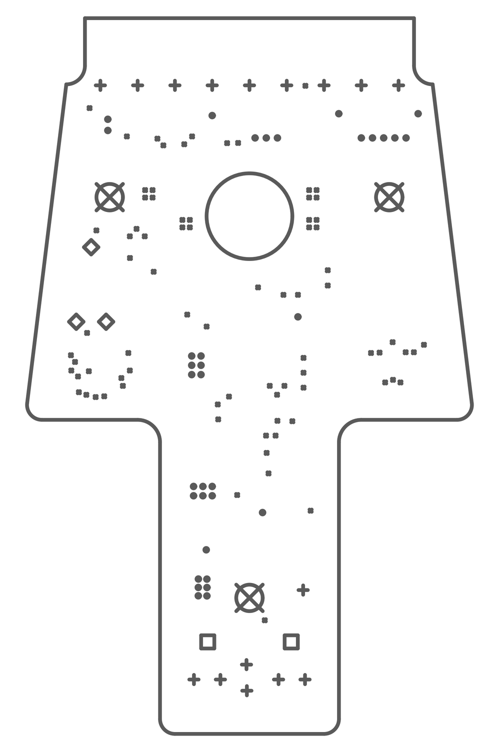
DESIGN ACTIVITY U.S. ARMY
COMBAT CAPABILITIES DEVELOPMENT COMMAND ARMAMENTS CENTER
PICATINNY ARSENAL, NJ 07806-5000

Fusion
Platform

SIZE	CAGE CODE	DWG. NO.
B	19200	PB-F-0210

SCALE: 1:1 | UNIT WEIGHT: | SHEET 1 OF 1





Drill Map:

- · 0.152mm / 0.0060'' (79 holes)
- · 0.254mm / 0.0100'' (34 holes)
- + 0.635mm / 0.0250'' (16 holes)
- □ 1.270mm / 0.0500'' (2 holes)
- ◇ 0.991mm / 0.0390'' (3 holes) (not plated)
- ⊠ 1.803mm / 0.0710'' (3 holes) (not plated)

LAYER STACKUP AND FILENAME

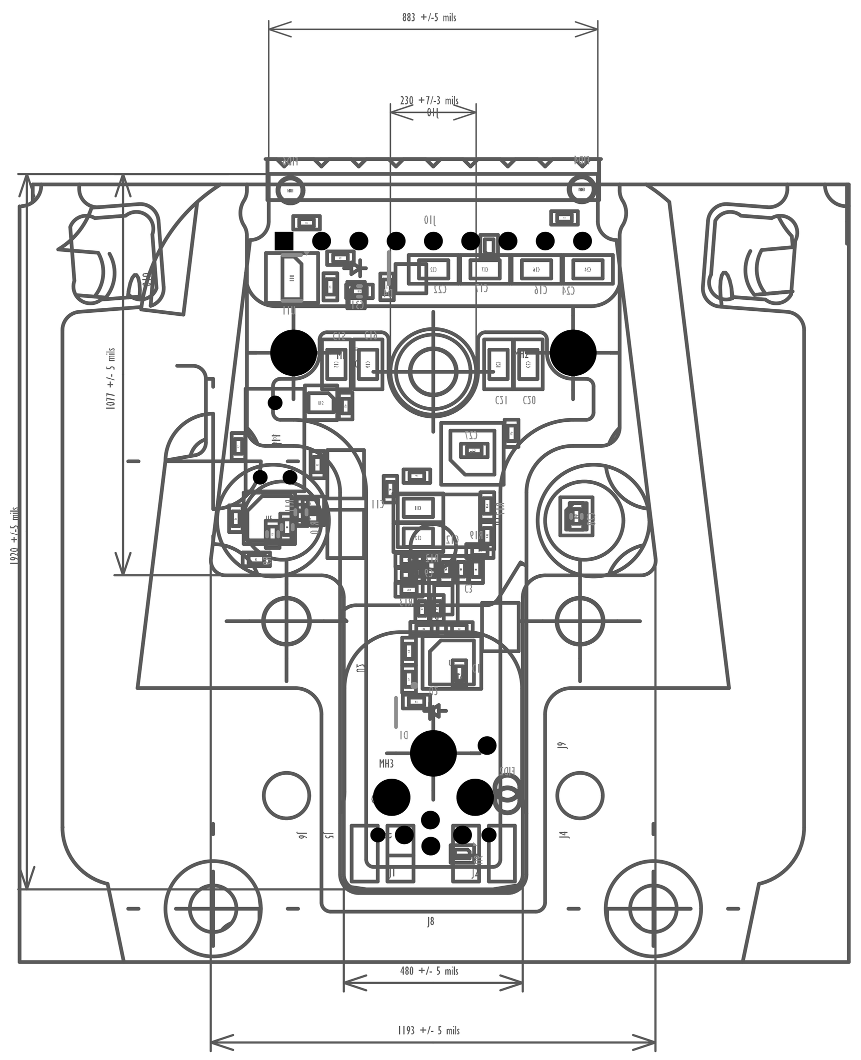
LAYER NAME	COPPER LAYER NUMBER	TYPE	FILENAME	THICKNESS (mils)	Er
FRONT SILKSCREEN	String	SILKSCREEN	100069026-001_Rev1-F_SILK.GBR	String	String
FRONT SOLDERMASK	String	SOLDERMASK	100069026-001_Rev1-F_MASK.GBR	1.00	4.50
FRONT COPPER	1	0.5 OZ COPPER	100069026-001_Rev1-F_CU.GBR	1.90	String
370HR PP106	String	PREPREG	String	2.30	3.81
370HR PP106	String	PREPREG	String	2.30	3.81
INNER 1 COPPER	2	1 OZ COPPER	100069026-001_Rev1-IN1_CU.GBR	1.40	String
370HR 1x7628	String	PREPREG	String	8.00	4.34
INNER 2 COPPER	3	1 OZ COPPER	100069026-001_Rev1-IN2_CU.GBR	1.40	String
370HR PP2113	String	PREPREG	String	3.90	4.09
370HR PP2113	String	PREPREG	String	3.90	4.09
INNER 3 COPPER	4	1 OZ COPPER	100069026-001_Rev1-IN3_CU.GBR	1.40	String
370HR 1x7628	String	PREPREG	String	8.00	4.34
INNER 4 COPPER	5	1 OZ COPPER	100069026-001_Rev1-IN4_CU.GBR	1.40	String
370HR PP2113	String	PREPREG	String	3.5	4.09
370HR PP2113	String	PREPREG	String	3.5	4.09
INNER 5 COPPER	6	1 OZ COPPER	100069026-001_Rev1-IN5_CU.GBR	1.40	String
370HR 1x7628	String	PREPREG	String	8.00	4.34
INNER 6 COPPER	7	1 OZ COPPER	100069026-001_Rev1-IN6_CU.GBR	1.40	String
370HR PP106	String	PREPREG	String	2.30	3.81
370HR PP106	String	PREPREG	String	2.30	3.81
BACK COPPER	8	0.5 OZ COPPER	100069026-001_Rev1-B_CU.GBR	1.90	String
BACK SOLDERMASK	String	SOLDERMASK	100069026-001_Rev1-B_MASK.GBR	1.00	4.50
BACK SILKSCREEN	String	SILKSCREEN	100069026-001_Rev1-B_SILK.GBR	String	String
			TOTAL THICKNESS	62.0 +/-5	
		BOARD OUTLINE	100069026-001_Rev1_EDGECUTS.GBR		
		PLATED THRU HOLES	100069026-001_Rev1_PTH.DRL		
		NON-PLATED THRU HOLES	100069026-001_Rev1_NPTH.DRL		

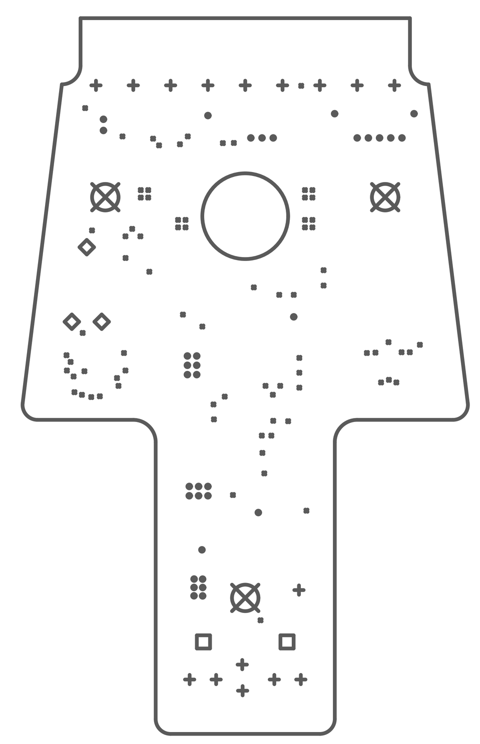
NOTE: FINISHED OUTER COPPER WEIGHT IS 1 OZ. THIS SHOULD BE 1/2 OZ BASE FOIL AND 1/2 OZ PLATING.

IMPEDANCE

TRACE LAYER	REF PLANE LAYER	IMPEDANCE (OHMS)	TRACE WIDTH (MILS)	TRACE SPACE (MILS)	TYPE
5	4, 6	90	5.4	6.8	DIFF COPLANAR WAVEGUIDE, 2 PLANES

NOTES: UNLESS OTHERWISE SPECIFIED
1: STANDARDS
1.1: THE PCB SHALL BE FABRICATED IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-6012 CLASS 2.
1.2: INTERPRET DIMENSIONS AND TOLERANCES IN ACCORDANCE WITH THE CURRENT REVISION OF ASME Y14.5M.
1.3: TEST IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-TM-650 2.4.22.
1.4: DO NOT SCALE TO DRAWING.
2: MATERIAL
2.1: THE PCB DIELECTRIC MATERIAL SHALL BE 370HR.
2.2: THICKNESSES OF CONDUCTOR AND DIELECTRIC SHALL MATCH THE STACKUP DRAWING.
3: FLATNESS
3.1: BOW AND TWIST OF ASSEMBLY SUB-PANEL OR SINGULATED PWB SHALL NOT EXCEED 0.75%.
3.2: TEST IN ACCORDANCE WITH CURRENT REVISION OF IPC-TM-650 2.4.22.
4: ETCH GEOMETRY.
4.1: MEASURE WIDTH FROM THE BASE OF THE METALLIZATION.
4.2: FINISHED LINE WIDTH AND TERMINAL AREA SHALL NOT DEVIATE FROM THE GERBER PATTERN IMAGE BY MORE THAN 20%.
5: SURFACE FINISH
5.1: FINAL FINISH SHALL BE IN ACCORDANCE WITH IPC-4552 AND SHALL BE ENIG.
6: HOLES
6.1: SEE HOLE TABLE FOR DIMENSIONS (AFTER PLATING WHERE APPLICABLE).
6.2: PLATING IN HOLES SHALL BE CONTINUOUS ELECTROLYTIC COPPER IN ACCORDANCE WITH IPC-6012, CLASS 2.
6.3: SEE DRILL CHART FOR FINISHED HOLE SIZE AND PLATING.
6.4: ALL HOLES SHALL BE LOCATED WITHIN 3 MIL OF TRUE POSITION AS SUPPLIED IN CAD DATA.
6.5: THE TOLERANCE FOR ALL HOLES SHALL BE AS INDICATED IN THE HOLE TABLE.
6.6: HOLES IN THE HOLE TABLE MARKED AS {dblquote}CAPPED/FILLED{dblquote} SHALL BE FILLED WITH A NON-CONDUCTIVE MATERIAL AND PLATED OVER.
7: SOLDERMASK
7.1: SOLDERMASK OVER BARE COPPER (SMOBC) ON PRIMARY AND SECONDARY SIDES USING SUPPLIED ARTWORK IN ACCORDANCE WITH CURRENT REVISION OF IPC-SM-840 TYPE B.
7.2: COLOR: GREEN.
7.3: LIQUID PHOTO-IMAGEABLE (LPI) 0.001 INCH TO 0.002 INCH THICKNESS, HALOGEN FREE.
7.4: NO BLEED-OUT ALLOWED OVER EXPOSED SMD PADS.
8: SILKSCREEN
8.1: SILKSCREEN PRIMARY AND SECONDARY SIDE WITH NON-CONDUCTIVE, NON-NUTRIENT EPOXY INK.
8.2: COLOR: WHITE.
8.3: ANY UNSPECIFIED STROKE WIDTH SHALL BE 5 MIL.
8.4: CLIP SILKSCREEN AWAY FROM ANY EXPOSED METAL.
9: FINISHING
9.1: REMOVE ALL BURRS AND BREAK SHARP EDGES.
10: IMPEDANCE (ALL TOLERANCED +/- 10%)
10.1: SEE IMPEDANCE REFERENCE TABLES.
10.2: VENDOR MAY ADJUST DESIGN GEOMETRIES UP TO +/-20% TO ACHIEVE TARGET IMPEDANCE. ADJUSTMENTS BEYOND 20% OF LINE WIDTH, SPACING OR DIELECTRIC THICKNESS SHALL REQUIRE APPROVAL FROM DRAPER ENGINEERING.
10.3: VENDOR MAY SUBSTITUTE SPECIFIC CORES OR PREPREGS TO ACHIEVE IMPEDANCE TARGETS AS LONG AS THE DIELECTRIC REMAINS UNCHANGED.
11: PANELIZED PWBS LAYOUT
11.1: IF THE PWB IS PANELIZED, THE VENDOR HAS THE FREEDOM TO MOVE INDIVIDUAL OR GROUPED PWBS TO ANY LOCATION ON THE PANEL TO EASE FABRICATION OR TEST.
12: NON-DESTRUCTIVE EVALUATION
12.1: ALL PWBS SHALL PASS 100% ELECTRICAL TEST USING SUPPLIED IPC-D-365 NETLIST IN ACCORDANCE WITH CURRENT REVISION OF IPC-9252, CLASS 2.
12.2: CERTIFICATE OF CONFORMANCE SHALL BE SUPPLIED WITH EACH SHIPMENT.
12.3: ALL BARE BOARDS THAT PASS ELECTRICAL TEST SHALL BE MARKED WITH WHITE EPOXY INK, IN ACCORDANCE WITH MIL-STD-130, WITH THE LETTERS 'ET'. MARKING SHALL NOT INTERFERE WITH ANY TRACE OR MOUNTING PAD.





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FRONT COPPER	1	0.5 OZ COPPER	100069026-001_Rev1-F_CU.GBR	1.90	String
370HR PP106	String	PREPREG	String	2.30	3.81
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INNER 1 COPPER	2	1 OZ COPPER	100069026-001_Rev1-IN1_CU.GBR	1.40	String
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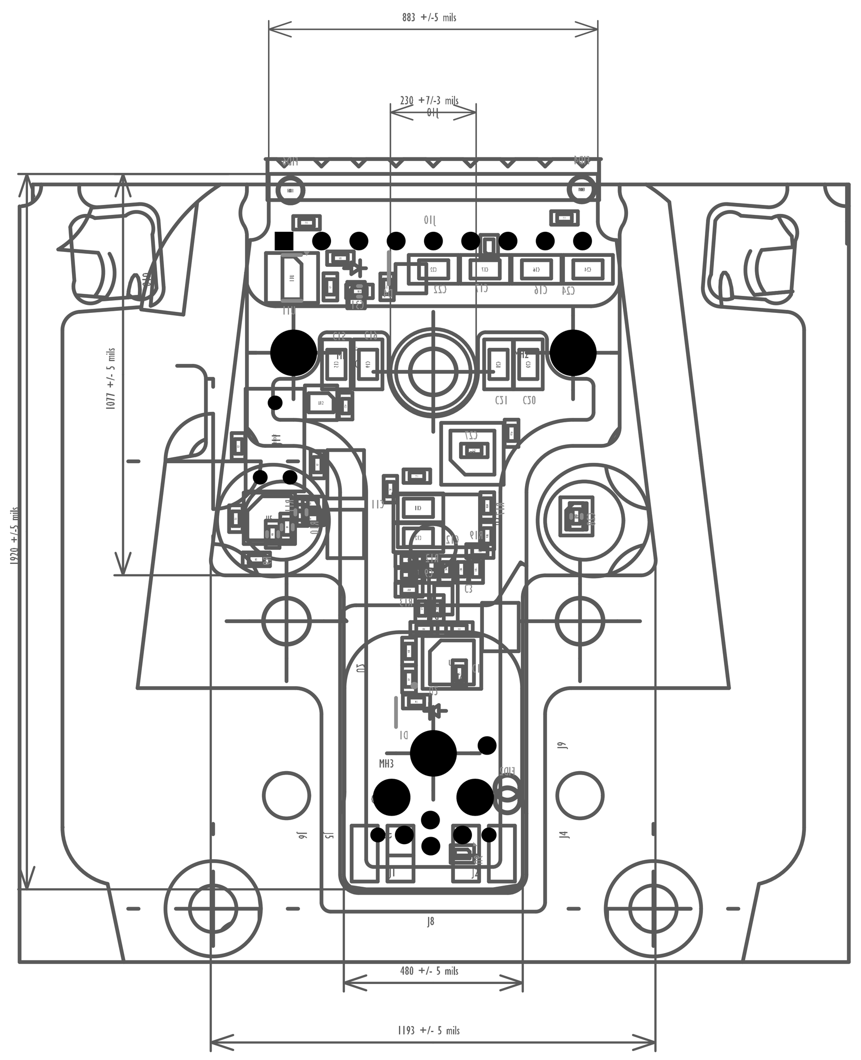

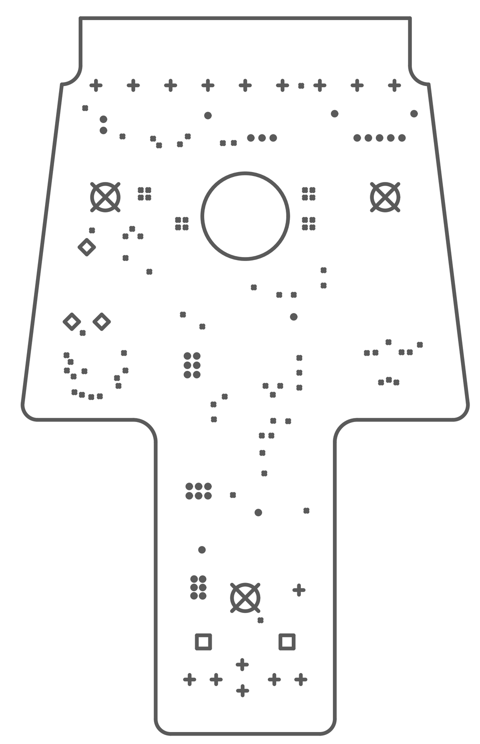

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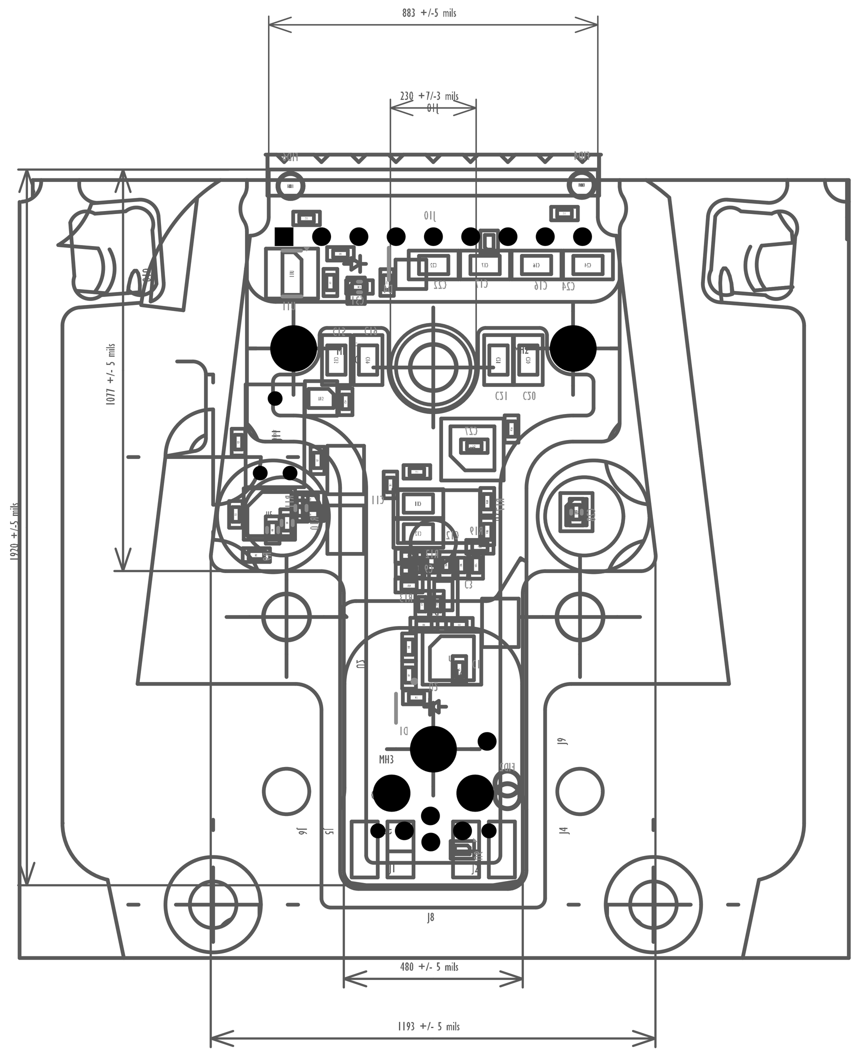
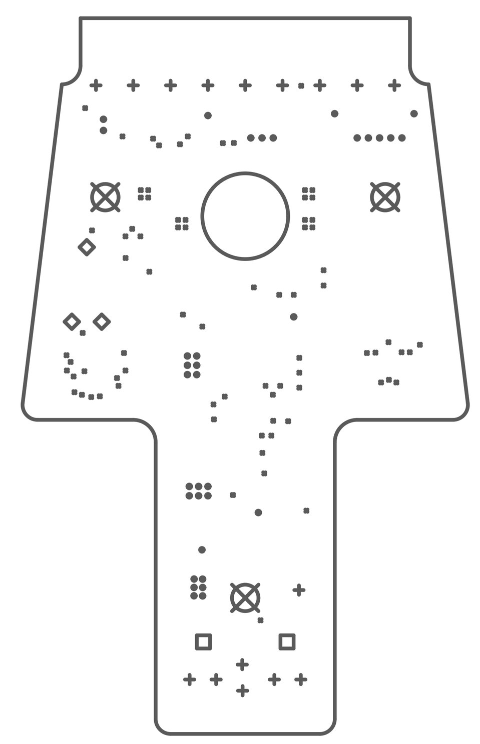
11.1: IF THE PWB IS PANELIZED, THE VENDOR HAS THE FREEDOM TO MOVE INDIVIDUAL OR GROUPED PWBS TO ANY LOCATION ON THE PANEL TO EASE FABRICATION OR TEST.
12: NON-DESTRUCTIVE EVALUATION
12.1: ALL PWBS SHALL PASS 100% ELECTRICAL TEST USING SUPPLIED IPC-D-365 NETLIST IN ACCORDANCE WITH CURRENT REVISION OF IPC-9252, CLASS 2.
12.2: CERTIFICATE OF CONFORMANCE SHALL BE SUPPLIED WITH EACH SHIPMENT.
12.3: ALL BARE BOARDS THAT PASS ELECTRICAL TEST SHALL BE MARKED WITH WHITE EPOXY INK, IN ACCORDANCE WITH MIL-STD-130, WITH THE LETTERS 'ET'. MARKING SHALL NOT INTERFERE WITH ANY TRACE OR MOUNTING PAD.





Drill Map:

- · 0.152mm / 0.0060'' (79 holes)
- · 0.254mm / 0.0100'' (34 holes)
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- ◇ 0.991mm / 0.0390'' (3 holes) (not plated)
- ⊠ 1.803mm / 0.0710'' (3 holes) (not plated)

LAYER STACKUP AND FILENAME

LAYER NAME	COPPER LAYER NUMBER	TYPE	FILENAME	THICKNESS (mils)	Er
FRONT SILKSCREEN	String	SILKSCREEN	100069026-001_Rev1-F_SILK.GBR	String	String
FRONT SOLDERMASK	String	SOLDERMASK	100069026-001_Rev1-F_MASK.GBR	1.00	4.50
FRONT COPPER	1	0.5 OZ COPPER	100069026-001_Rev1-F_CU.GBR	1.90	String
370HR PP106	String	PREPREG	String	2.30	3.81
370HR PP106	String	PREPREG	String	2.30	3.81
INNER 1 COPPER	2	1 OZ COPPER	100069026-001_Rev1-IN1_CU.GBR	1.40	String
370HR 1x7628	String	PREPREG	String	8.00	4.34
INNER 2 COPPER	3	1 OZ COPPER	100069026-001_Rev1-IN2_CU.GBR	1.40	String
370HR PP2113	String	PREPREG	String	3.90	4.09
370HR PP2113	String	PREPREG	String	3.90	4.09
INNER 3 COPPER	4	1 OZ COPPER	100069026-001_Rev1-IN3_CU.GBR	1.40	String
370HR 1x7628	String	PREPREG	String	8.00	4.34
INNER 4 COPPER	5	1 OZ COPPER	100069026-001_Rev1-IN4_CU.GBR	1.40	String
370HR PP2113	String	PREPREG	String	3.5	4.09
370HR PP2113	String	PREPREG	String	3.5	4.09
INNER 5 COPPER	6	1 OZ COPPER	100069026-001_Rev1-IN5_CU.GBR	1.40	String
370HR 1x7628	String	PREPREG	String	8.00	4.34
INNER 6 COPPER	7	1 OZ COPPER	100069026-001_Rev1-IN6_CU.GBR	1.40	String
370HR PP106	String	PREPREG	String	2.30	3.81
370HR PP106	String	PREPREG	String	2.30	3.81
BACK COPPER	8	0.5 OZ COPPER	100069026-001_Rev1-B_CU.GBR	1.90	String
BACK SOLDERMASK	String	SOLDERMASK	100069026-001_Rev1-B_MASK.GBR	1.00	4.50
BACK SILKSCREEN	String	SILKSCREEN	100069026-001_Rev1-B_SILK.GBR	String	String
			TOTAL THICKNESS	62.0 +/-5	
		BOARD OUTLINE	100069026-001_Rev1_EDGECUTS.GBR		
		PLATED THRU HOLES	100069026-001_Rev1_PTH.DRL		
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NOTE: FINISHED OUTER COPPER WEIGHT IS 1 OZ. THIS SHOULD BE 1/2 OZ BASE FOIL AND 1/2 OZ PLATING.

IMPEDANCE

TRACE LAYER	REF PLANE LAYER	IMPEDANCE (OHMS)	TRACE WIDTH (MILS)	TRACE SPACE (MILS)	TYPE
5	4, 6	90	5.4	6.8	DIFF COPLANAR WAVEGUIDE, 2 PLANES

NOTES: UNLESS OTHERWISE SPECIFIED
1: STANDARDS
1.1: THE PCB SHALL BE FABRICATED IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-6012 CLASS 2.
1.2: INTERPRET DIMENSIONS AND TOLERANCES IN ACCORDANCE WITH THE CURRENT REVISION OF ASME Y14.5M.
1.3: TEST IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-TM-650 2.4.22.
1.4: DO NOT SCALE TO DRAWING.
2: MATERIAL
2.1: THE PCB DIELECTRIC MATERIAL SHALL BE 370HR.
2.2: THICKNESSES OF CONDUCTOR AND DIELECTRIC SHALL MATCH THE STACKUP DRAWING.
3: FLATNESS
3.1: BOW AND TWIST OF ASSEMBLY SUB-PANEL OR SINGULATED PWB SHALL NOT EXCEED 0.75%.
3.2: TEST IN ACCORDANCE WITH CURRENT REVISION OF IPC-TM-650 2.4.22.
4: ETCH GEOMETRY.
4.1: MEASURE WIDTH FROM THE BASE OF THE METALLIZATION.
4.2: FINISHED LINE WIDTH AND TERMINAL AREA SHALL NOT DEVIATE FROM THE GERBER PATTERN IMAGE BY MORE THAN 20%.
5: SURFACE FINISH
5.1: FINAL FINISH SHALL BE IN ACCORDANCE WITH IPC-4552 AND SHALL BE ENIG.
6: HOLES
6.1: SEE HOLE TABLE FOR DIMENSIONS (AFTER PLATING WHERE APPLICABLE).
6.2: PLATING IN HOLES SHALL BE CONTINUOUS ELECTROLYTIC COPPER IN ACCORDANCE WITH IPC-6012, CLASS 2.
6.3: SEE DRILL CHART FOR FINISHED HOLE SIZE AND PLATING.
6.4: ALL HOLES SHALL BE LOCATED WITHIN 3 MIL OF TRUE POSITION AS SUPPLIED IN CAD DATA.
6.5: THE TOLERANCE FOR ALL HOLES SHALL BE AS INDICATED IN THE HOLE TABLE.
6.6: HOLES IN THE HOLE TABLE MARKED AS {dblquote}CAPPED/FILLED{dblquote} SHALL BE FILLED WITH A NON-CONDUCTIVE MATERIAL AND PLATED OVER.
7: SOLDERMASK
7.1: SOLDERMASK OVER BARE COPPER (SMOBC) ON PRIMARY AND SECONDARY SIDES USING SUPPLIED ARTWORK IN ACCORDANCE WITH CURRENT REVISION OF IPC-SM-840 TYPE B.
7.2: COLOR: GREEN.
7.3: LIQUID PHOTO-IMAGEABLE (LPI) 0.001 INCH TO 0.002 INCH THICKNESS, HALOGEN FREE.
7.4: NO BLEED-OUT ALLOWED OVER EXPOSED SMD PADS.
8: SILKSCREEN
8.1: SILKSCREEN PRIMARY AND SECONDARY SIDE WITH NON-CONDUCTIVE, NON-NUTRIENT EPOXY INK.
8.2: COLOR: WHITE.
8.3: ANY UNSPECIFIED STROKE WIDTH SHALL BE 5 MIL.
8.4: CLIP SILKSCREEN AWAY FROM ANY EXPOSED METAL.
9: FINISHING
9.1: REMOVE ALL BURRS AND BREAK SHARP EDGES.
10: IMPEDANCE (ALL TOLERANCED +/- 10%)
10.1: SEE IMPEDANCE REFERENCE TABLES.
10.2: VENDOR MAY ADJUST DESIGN GEOMETRIES UP TO +/-20% TO ACHIEVE TARGET IMPEDANCE. ADJUSTMENTS BEYOND 20% OF LINE WIDTH, SPACING OR DIELECTRIC THICKNESS SHALL REQUIRE APPROVAL FROM DRAPER ENGINEERING.
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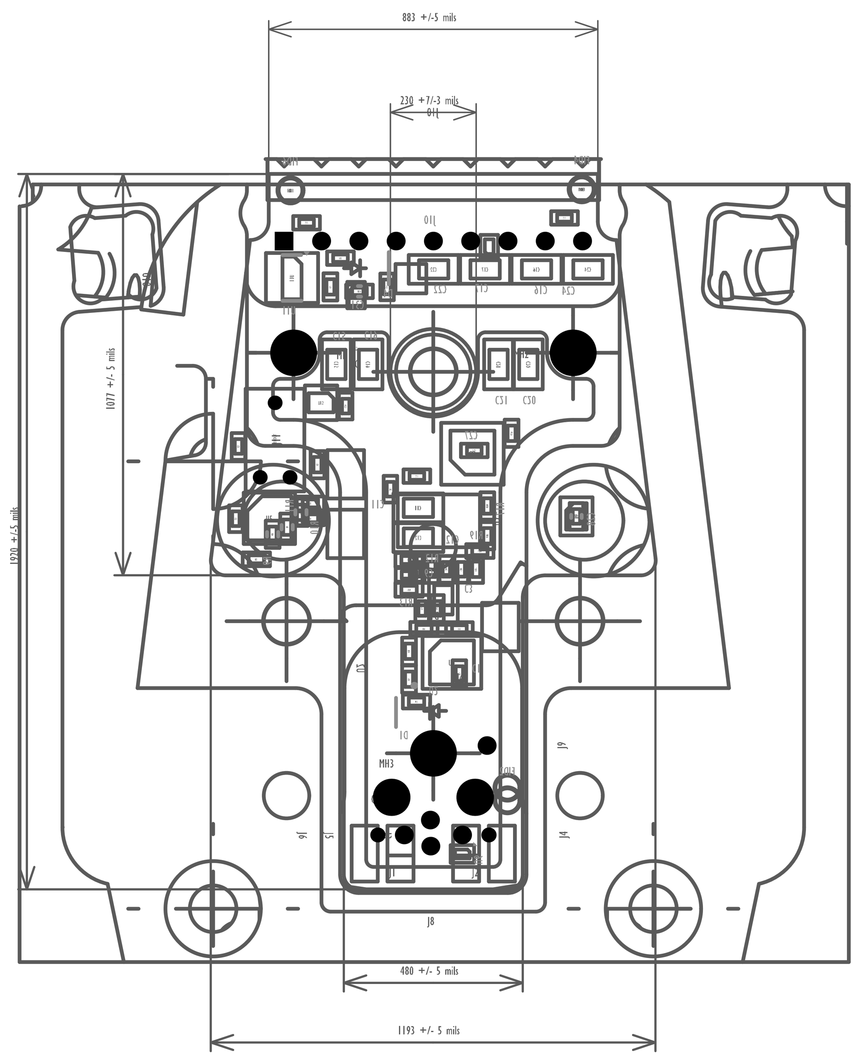

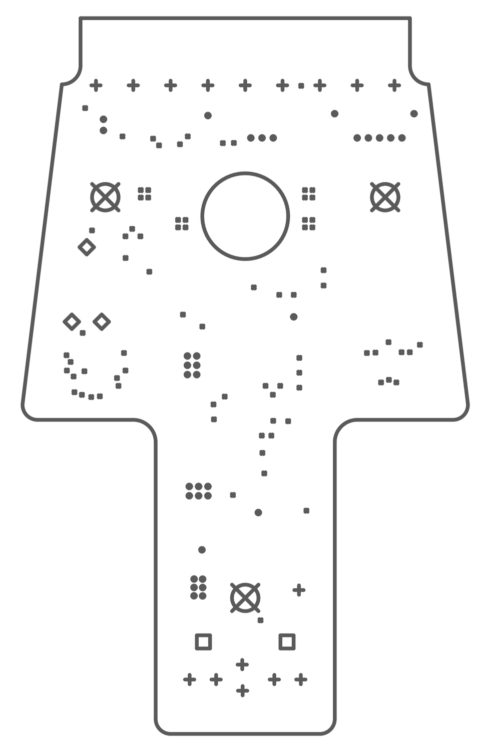

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LAYER STACKUP AND FILENAME

LAYER NAME	COPPER LAYER NUMBER	TYPE	FILENAME	THICKNESS (mils)	Er
FRONT SILKSCREEN	String	SILKSCREEN	100069026-001_Rev1-F_SILK.GBR	String	String
FRONT SOLDERMASK	String	SOLDERMASK	100069026-001_Rev1-F_MASK.GBR	1.00	4.50
FRONT COPPER	1	0.5 OZ COPPER	100069026-001_Rev1-F_CU.GBR	1.90	String
370HR PP106	String	PREPREG	String	2.30	3.81
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1.3: TEST IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-TM-650 2.4.22.
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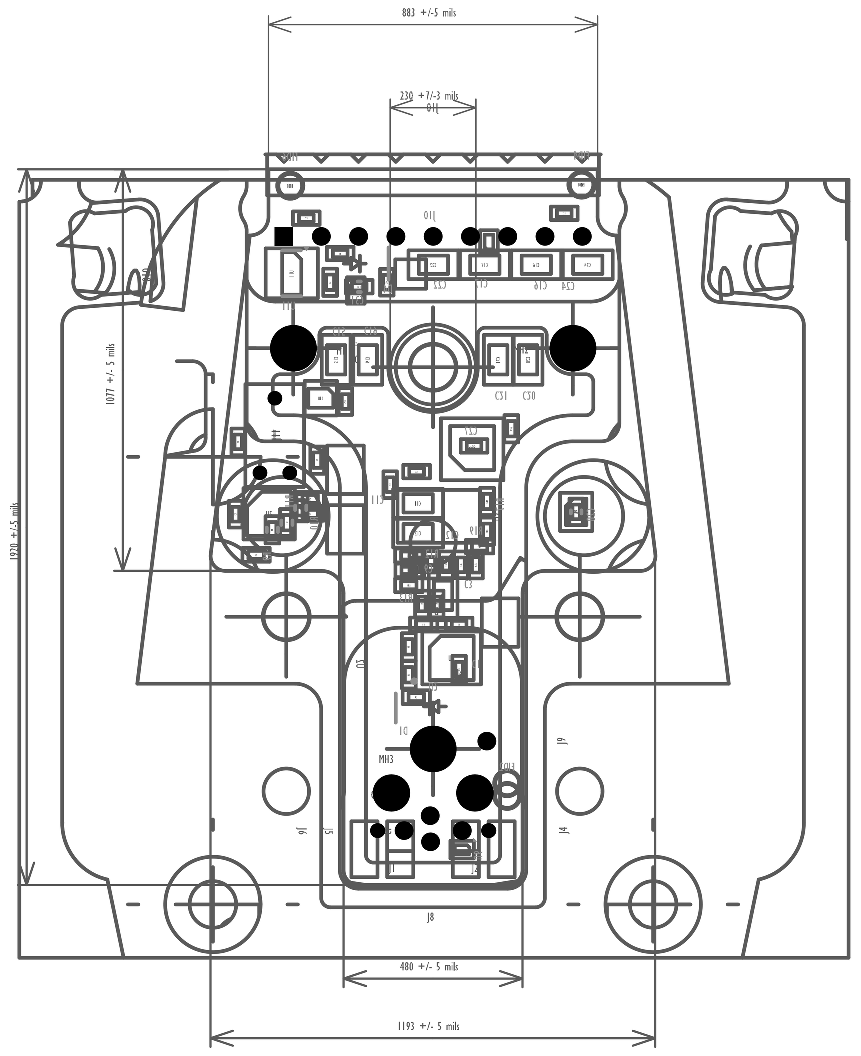

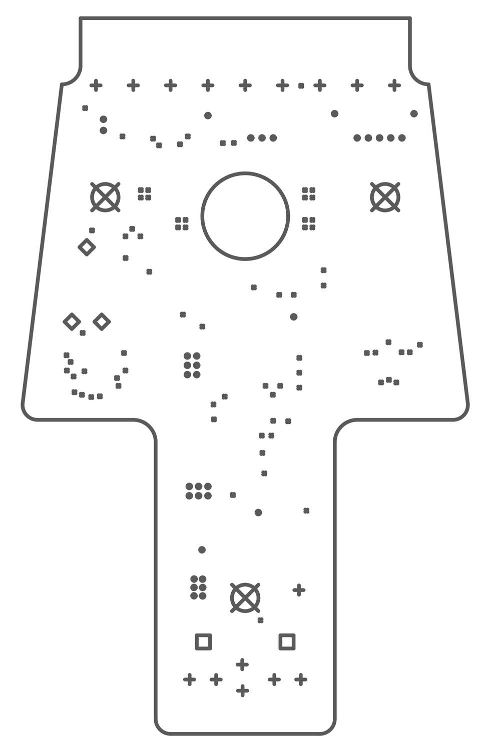

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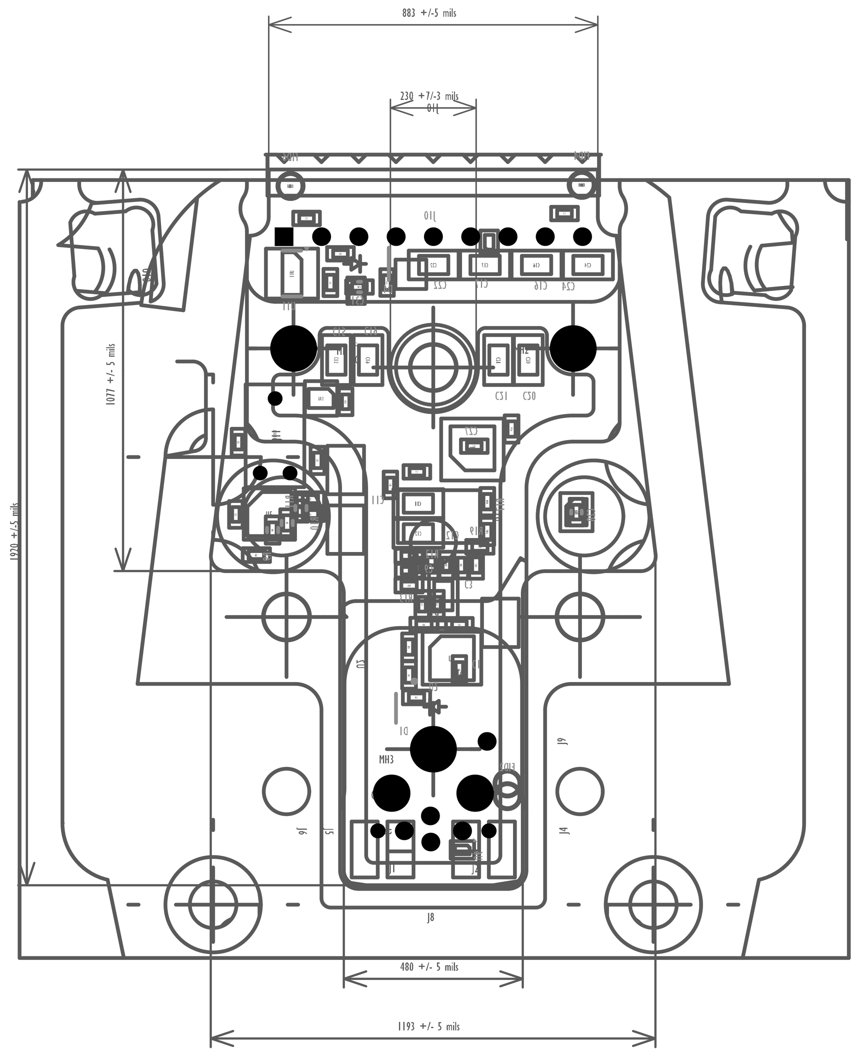

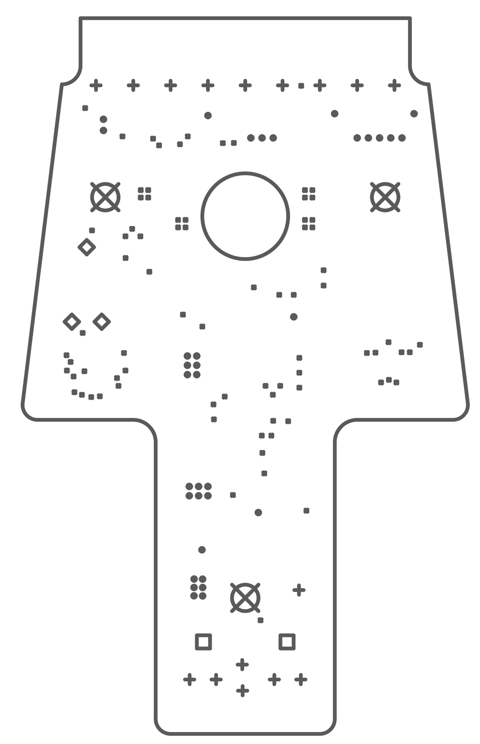

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- · 0.254mm / 0.0100'' (34 holes)
- + 0.635mm / 0.0250'' (16 holes)
- □ 1.270mm / 0.0500'' (2 holes)
- ◇ 0.991mm / 0.0390'' (3 holes) (not plated)
- ⊠ 1.803mm / 0.0710'' (3 holes) (not plated)

LAYER STACKUP AND FILENAME

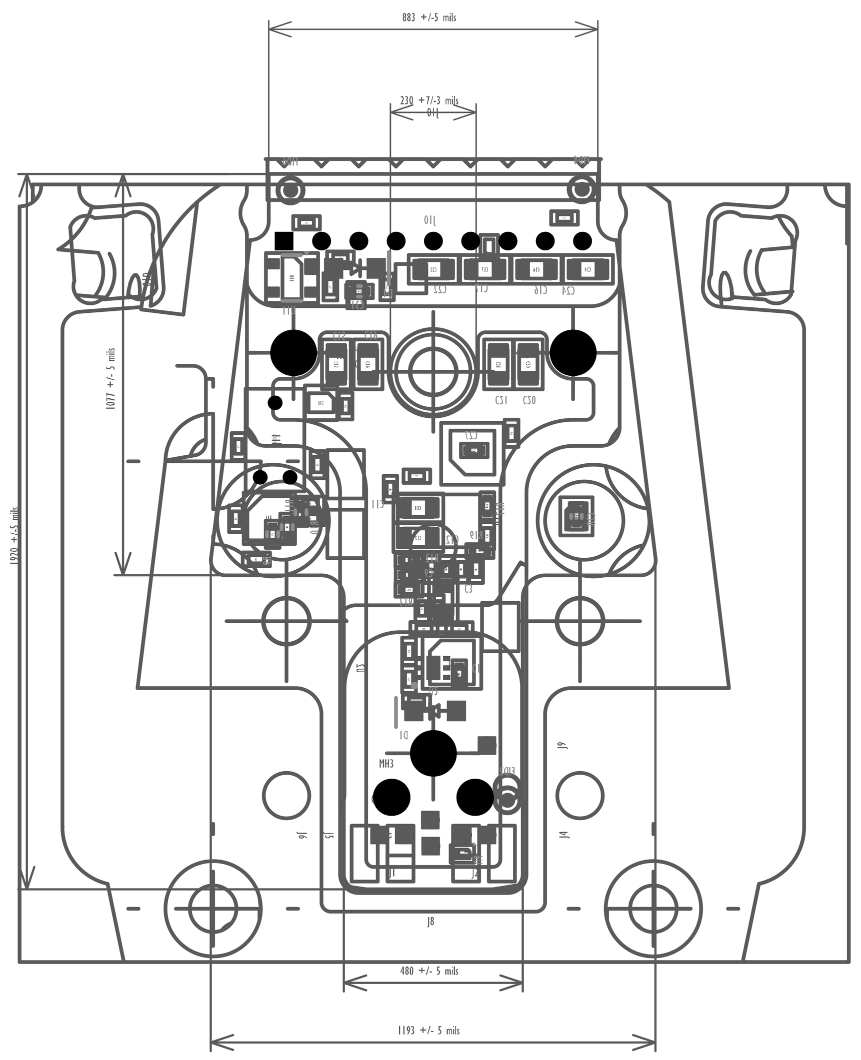
LAYER NAME	COPPER LAYER NUMBER	TYPE	FILENAME	THICKNESS (mils)	Er
FRONT SILKSCREEN	String	SILKSCREEN	100069026-001_Rev1-F_SILK.GBR	String	String
FRONT SOLDERMASK	String	SOLDERMASK	100069026-001_Rev1-F_MASK.GBR	1.00	4.50
FRONT COPPER	1	0.5 OZ COPPER	100069026-001_Rev1-F_CU.GBR	1.90	String
370HR PP106	String	PREPREG	String	2.30	3.81
370HR PP106	String	PREPREG	String	2.30	3.81
INNER 1 COPPER	2	1 OZ COPPER	100069026-001_Rev1-IN1_CU.GBR	1.40	String
370HR 1x7628	String	PREPREG	String	8.00	4.34
INNER 2 COPPER	3	1 OZ COPPER	100069026-001_Rev1-IN2_CU.GBR	1.40	String
370HR PP2113	String	PREPREG	String	3.90	4.09
370HR PP2113	String	PREPREG	String	3.90	4.09
INNER 3 COPPER	4	1 OZ COPPER	100069026-001_Rev1-IN3_CU.GBR	1.40	String
370HR 1x7628	String	PREPREG	String	8.00	4.34
INNER 4 COPPER	5	1 OZ COPPER	100069026-001_Rev1-IN4_CU.GBR	1.40	String
370HR PP2113	String	PREPREG	String	3.5	4.09
370HR PP2113	String	PREPREG	String	3.5	4.09
INNER 5 COPPER	6	1 OZ COPPER	100069026-001_Rev1-IN5_CU.GBR	1.40	String
370HR 1x7628	String	PREPREG	String	8.00	4.34
INNER 6 COPPER	7	1 OZ COPPER	100069026-001_Rev1-IN6_CU.GBR	1.40	String
370HR PP106	String	PREPREG	String	2.30	3.81
370HR PP106	String	PREPREG	String	2.30	3.81
BACK COPPER	8	0.5 OZ COPPER	100069026-001_Rev1-B_CU.GBR	1.90	String
BACK SOLDERMASK	String	SOLDERMASK	100069026-001_Rev1-B_MASK.GBR	1.00	4.50
BACK SILKSCREEN	String	SILKSCREEN	100069026-001_Rev1-B_SILK.GBR	String	String
			TOTAL THICKNESS	62.0 +/-5	
		BOARD OUTLINE	100069026-001_Rev1_EDGECUTS.GBR		
		PLATED THRU HOLES	100069026-001_Rev1_PTH.DRL		
		NON-PLATED THRU HOLES	100069026-001_Rev1_NPTH.DRL		

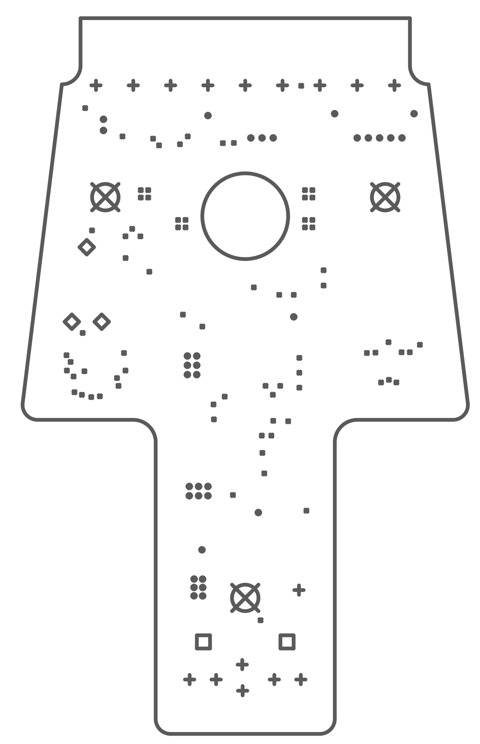
NOTE: FINISHED OUTER COPPER WEIGHT IS 1 OZ. THIS SHOULD BE 1/2 OZ BASE FOIL AND 1/2 OZ PLATING.

IMPEDANCE

TRACE LAYER	REF PLANE LAYER	IMPEDANCE (OHMS)	TRACE WIDTH (MILS)	TRACE SPACE (MILS)	TYPE
5	4, 6	90	5.4	6.8	DIFF COPLANAR WAVEGUIDE, 2 PLANES

NOTES: UNLESS OTHERWISE SPECIFIED
1: STANDARDS
1.1: THE PCB SHALL BE FABRICATED IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-6012 CLASS 2.
1.2: INTERPRET DIMENSIONS AND TOLERANCES IN ACCORDANCE WITH THE CURRENT REVISION OF ASME Y14.5M.
1.3: TEST IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-TM-650 2.4.22.
1.4: DO NOT SCALE TO DRAWING.
2: MATERIAL
2.1: THE PCB DIELECTRIC MATERIAL SHALL BE 370HR.
2.2: THICKNESSES OF CONDUCTOR AND DIELECTRIC SHALL MATCH THE STACKUP DRAWING.
3: FLATNESS
3.1: BOW AND TWIST OF ASSEMBLY SUB-PANEL OR SINGULATED PWB SHALL NOT EXCEED 0.75%.
3.2: TEST IN ACCORDANCE WITH CURRENT REVISION OF IPC-TM-650 2.4.22.
4: ETCH GEOMETRY.
4.1: MEASURE WIDTH FROM THE BASE OF THE METALLIZATION.
4.2: FINISHED LINE WIDTH AND TERMINAL AREA SHALL NOT DEVIATE FROM THE GERBER PATTERN IMAGE BY MORE THAN 20%.
5: SURFACE FINISH
5.1: FINAL FINISH SHALL BE IN ACCORDANCE WITH IPC-4552 AND SHALL BE ENIG.
6: HOLES
6.1: SEE HOLE TABLE FOR DIMENSIONS (AFTER PLATING WHERE APPLICABLE).
6.2: PLATING IN HOLES SHALL BE CONTINUOUS ELECTROLYTIC COPPER IN ACCORDANCE WITH IPC-6012, CLASS 2.
6.3: SEE DRILL CHART FOR FINISHED HOLE SIZE AND PLATING.
6.4: ALL HOLES SHALL BE LOCATED WITHIN 3 MIL OF TRUE POSITION AS SUPPLIED IN CAD DATA.
6.5: THE TOLERANCE FOR ALL HOLES SHALL BE AS INDICATED IN THE HOLE TABLE.
6.6: HOLES IN THE HOLE TABLE MARKED AS {dblquote}CAPPED/FILLED{dblquote} SHALL BE FILLED WITH A NON-CONDUCTIVE MATERIAL AND PLATED OVER.
7: SOLDERMASK
7.1: SOLDERMASK OVER BARE COPPER (SMOBC) ON PRIMARY AND SECONDARY SIDES USING SUPPLIED ARTWORK IN ACCORDANCE WITH CURRENT REVISION OF IPC-SM-840 TYPE B.
7.2: COLOR: GREEN.
7.3: LIQUID PHOTO-IMAGEABLE (LPI) 0.001 INCH TO 0.002 INCH THICKNESS, HALOGEN FREE.
7.4: NO BLEED-OUT ALLOWED OVER EXPOSED SMD PADS.
8: SILKSCREEN
8.1: SILKSCREEN PRIMARY AND SECONDARY SIDE WITH NON-CONDUCTIVE, NON-NUTRIENT EPOXY INK.
8.2: COLOR: WHITE.
8.3: ANY UNSPECIFIED STROKE WIDTH SHALL BE 5 MIL.
8.4: CLIP SILKSCREEN AWAY FROM ANY EXPOSED METAL.
9: FINISHING
9.1: REMOVE ALL BURRS AND BREAK SHARP EDGES.
10: IMPEDANCE (ALL TOLERANCED +/- 10%)
10.1: SEE IMPEDANCE REFERENCE TABLES.
10.2: VENDOR MAY ADJUST DESIGN GEOMETRIES UP TO +/-20% TO ACHIEVE TARGET IMPEDANCE. ADJUSTMENTS BEYOND 20% OF LINE WIDTH, SPACING OR DIELECTRIC THICKNESS SHALL REQUIRE APPROVAL FROM DRAPER ENGINEERING.
10.3: VENDOR MAY SUBSTITUTE SPECIFIC CORES OR PREPREGS TO ACHIEVE IMPEDANCE TARGETS AS LONG AS THE DIELECTRIC REMAINS UNCHANGED.
11: PANELIZED PWBS LAYOUT
11.1: IF THE PWB IS PANELIZED, THE VENDOR HAS THE FREEDOM TO MOVE INDIVIDUAL OR GROUPED PWBS TO ANY LOCATION ON THE PANEL TO EASE FABRICATION OR TEST.
12: NON-DESTRUCTIVE EVALUATION
12.1: ALL PWBS SHALL PASS 100% ELECTRICAL TEST USING SUPPLIED IPC-D-365 NETLIST IN ACCORDANCE WITH CURRENT REVISION OF IPC-9252, CLASS 2.
12.2: CERTIFICATE OF CONFORMANCE SHALL BE SUPPLIED WITH EACH SHIPMENT.
12.3: ALL BARE BOARDS THAT PASS ELECTRICAL TEST SHALL BE MARKED WITH WHITE EPOXY INK, IN ACCORDANCE WITH MIL-STD-130, WITH THE LETTERS 'ET'. MARKING SHALL NOT INTERFERE WITH ANY TRACE OR MOUNTING PAD.





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